Assignment 4: Gradient Descent Algorithm

Find the local minima of $y = (x + 3)^2$ starting from $x = 2$.

In [1]:
```
import matplotlib.pyplot as plt
import numpy as np
```

Function and its derivative

In [2]:
```
def f(x):
    return (x + 3) ** 2


def df(x):
    return 2 * (x + 3)
```

Gradient descent implementation

In [3]:
```
def gradient_descent(start_x, learning_rate=0.1, iterations=100,
tolerance=1e-6):
    x = start_x
    history = [x]
    for _ in range(iterations):
        grad = df(x)
        new_x = x - learning_rate * grad
        history.append(new_x)
        if abs(new_x - x) < tolerance:
            x = new_x
            break
        x = new_x
    return x, history
```

Run from x = 2

In [4]:
```
start_x = 2
learning_rate = 0.1
minima_x, history = gradient_descent(start_x, learning_rate)

print(f"Converged after {len(history) - 1} iterations")
print(f"Local minima at x = {minima_x:.6f}, y = {f(minima_x):.6f}")
print("Expected minima: x = -3, y = 0")
```

```
Converged after 63 iterations
Local minima at x = -2.999996, y = 0.000000
Expected minima: x = -3, y = 0
```

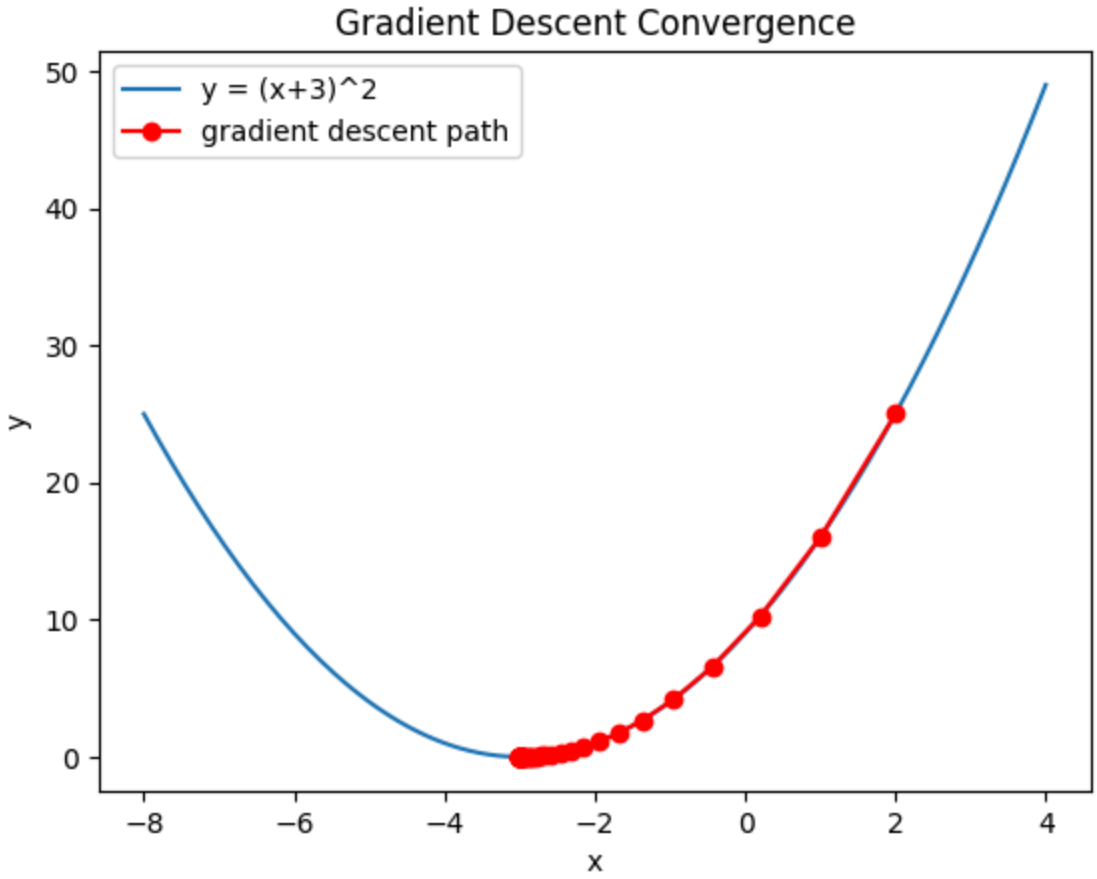
In [5]:
```
import pandas as pd
pd.DataFrame({"iteration": range(len(history)), "x": history, "y": [f(x)
for x in history]}).head(15)
```

Out[5]:

	iteration	x	y
0	0	2.000000	25.000000
1	1	1.000000	16.000000
2	2	0.200000	10.240000
3	3	-0.440000	6.553600
4	4	-0.952000	4.194304
5	5	-1.361600	2.684355
6	6	-1.689280	1.717987
7	7	-1.951424	1.099512
8	8	-2.161139	0.703687
9	9	-2.328911	0.450360
10	10	-2.463129	0.288230
11	11	-2.570503	0.184467
12	12	-2.656403	0.118059
13	13	-2.725122	0.075558
14	14	-2.780098	0.048357

Visualize convergence

In [6]:
```
xs = np.linspace(-8, 4, 200)
plt.plot(xs, f(xs), label="y = (x+3)^2")
plt.plot(history, [f(x) for x in history], "o-", color="red",
label="gradient descent path")
plt.xlabel("x")
plt.ylabel("y")
plt.legend()
plt.title("Gradient Descent Convergence")
plt.show()
```


Gradient Descent Convergence
y = (x+3)^2
gradient descent path
50
40
30
20
10
0
y
−8
−6
−4
−2
0
2
4
x